

# ABIR PALEKAR

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## PROFESSIONAL SUMMARY

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Aspiring high school junior with a strong interest in engineering and business. Passionate about exploring the engineering field through technical based courses and extracurricular activities whilst running a business, helping me understand how to manage finances, investments, and capital. Driven by the goal of bridging the gap between innovative design and financial viability to lead high-impact projects.

## EDUCATION

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### Naperville Central High School

Aug 2023 - Present

PSAT: 1440/1440 — SAT: 1500/1600 — ACT: 34/36

Relevant Coursework: AP Biology, AP Calculus, AP Computer Science, AP Physics 1, AP Language and Composition, Honors Chemistry, Cybersecurity

## SKILLS

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|-------------------------|-----------------------------------------------------------------------------------------------|
| <b>Technical Skills</b> | Supply Chain Management, Statistical Analysis, Version/Quality Control, Online Business Tools |
| <b>Soft Skills</b>      | Critical Thinking, Time Management, Team Coordination, Strategic Planning                     |
| <b>Coding</b>           | Python, HTML, Java, ClaudeCode                                                                |

## EXPERIENCE

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### Co-Founder

Jan 2026 - Present

PulseCore Electronics

*Naperville, IL*

- Achieved a 60% net profit margin by managing a \$50k inventory budget and leveraging data-driven arbitrage to scale sales across Amazon FBA and eBay, delivering a to a diverse customer base across the entire United States.
- Led end-to-end hardware restoration for memory and consumer electronics, converting bulk-sourced components into retail-grade assets with high-velocity sell-through.
- Developed a lean sourcing and fulfillment pipeline in a two-person partnership, overseeing 100% of procurement, technical QC, and multi-channel logistics.

### Computer Science Teacher

Aug 2025 - Dec 2025

Best Brains

*Naperville, IL*

- Educated 20+ elementary students in foundational computer science by translating complex logic into interactive, age-appropriate lesson plans.
- Improved student engagement and technical literacy by developing simplified curriculum focused on computational thinking and basic coding principles.

### Senior Marketer Intern

Jun 2025 - Nov 2025

Loopless LLC

*Northbrook, IL*

- Accelerated user growth for a high-traction digital health startup by producing viral short-form content that drove significant app downloads and audience engagement.
- Led content strategy for CBT-based wellness tools, translating psychological concepts into high-impact social media campaigns that increased brand visibility and retention.

### Chief Financial Officer

May 2025 - Present

LifeFlow (Startup)

*Naperville, IL*

- Architected the financial infrastructure for a very early-stage startup, developing comprehensive budget models and cash-flow projections to guide initial seed-stage strategy.

## Chess Instructor

Elite Chess

March 2024 - March 2025

*Aurora, IL*

- Instructed over 100 students on chess fundamentals and advanced tactics, tailoring lessons to help players of all skill levels improve.
- Led group coaching sessions for large classes, focusing on tournament preparation, sportsmanship, and sharpening analytical skills.

## VOLUNTEERING

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**Language Teacher** - Volunteered as a Marathi Language Instructor for 30+ children at Maharashtra Mandal Chicago, designing fun and engaging lessons to promote cultural literacy.

**Computer Science Teaching Assistant** - Represented the NCHS CS Club by teaching fundamental programming concepts and problem-solving to students at Kennedy Junior High. Through the CodeBytes program, I delivered interactive lessons and provided hands-on mentorship to simplify complex technical topics and inspire a lasting interest in STEM.

**Chess Instructor @ NCHS** - Supported the NCHS Chess program by teaching foundational and advanced chess concepts to elementary and middle school students. Played a key role in building program momentum and community engagement through personalized coaching and tournament mentorship.

**USCF Tournament Director** - Served as Chief Assistant Tournament Director for multiple USCF-sanctioned events, overseeing over 100 students while managing fair play, score reporting, and logistics in a high-stakes competitive environment.

## EXTRA-CURRICULAR ACTIVITIES

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### Varsity Volleyball

Naperville Central High School

Mar 2025 - Present

*Naperville, IL*

- Compete as a Varsity Volleyball athlete for NCHS, dedicating 15+ hours weekly to training and team strategy while maintaining a high academic workload.

### Redhawk ELITE

Naperville Central High School

Aug 2025 - Present

*Naperville, IL*

- Nominated for Redhawk ELITE, a coach-nominated leadership program for high-performing student-athletes, focusing on developing advanced leadership skills and driving cultural growth within the varsity athletic program.

### Spanish Honors Society

Naperville Central High School

April 2025 - Present

*Naperville, IL*

- Honors Society that recognizes students for outstanding academic achievement in the Spanish language and culture.

### Chess Team

Naperville Central High School

Aug 2023 - Present

*Naperville, IL*

- Compete as a three-year Varsity Chess starter and two-year Team Captain, securing a 6th-place finish at the 2024 IHSA State Championships as well as leading the team to three consecutive top-30 state rankings as a 3x State Qualifier.

### Computer Science Club

Naperville Central High School

Aug 2024 - Present

*Naperville, IL*

- Member of the CS Club, leading community outreach by teaching programming to junior high students through the CodeBytes program.

### Psychology Club

Naperville Central High School

Aug 2025 - Present

*Naperville, IL*

- Meet with other students to discuss and learn about different psychology topics.

### **Hawk TV**

Naperville Central High School

Jan 2026 - Present

*Naperville, IL*

- Serve as an video editor for Hawk TV, producing monthly news broadcasts for a student body of more than 2000 students, ensuring high-quality video content through precise editing and storytelling.

### **Spanish Club**

Naperville Central High School

Aug 2023 - Present

*Naperville, IL*

- Engage as a member of Spanish Club, participating in cultural activities and group discussions to improve language fluency and appreciation for culture.

### **Asian Student Association**

Naperville Central High School

Oct 2025 - Present

*Naperville, IL*

- Active member of ASA, collaborating on cultural events and community initiatives to promote awareness and celebrate Asian heritage within the school.

## **PROJECTS**

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### **Real-Time World Cup Match Tracker**

Personal Project

June 2026

*Embedded Systems, API Integration, Circuit Design*

- Built an Arduino-based embedded system displaying live World Cup scores and match clocks on an LCD display.
- Wrote custom C++ firmware for display control, shift register I/O expansion, and real-time serial communication.
- Developed a Python backend scraping live match data from ESPN as well as API integration.
- Implemented automated match detection with countdown timers and live clock synchronization.

### **AI Gesture-Controlled Desktop Assistant**

Personal Project

June 2026

*Python, OpenCV, MediaPipe, Google API, PyAutoGUI*

- Built a real-time AI desktop assistant enabling hands-free computer control through voice commands and hand gesture tracking. Implemented computer vision-based cursor control with adaptive smoothing for precise mouse movement, clicking, and dragging using MediaPipe hand landmarks.
- Integrated Google Gemini API to enable natural language interaction for conversational responses and system automation, including web navigation, tab control, volume adjustment, and typing mode. Developed voice recognition features for executing system commands and dictation.
- Enabled toggleable control modes for gesture-based input and AI-driven interaction, combining computer vision, speech recognition, and LLM-based reasoning into a unified desktop assistant.

### **ASL Gesture-Controlled Drone Interface**

Personal Project

June 2026

*Python, OpenCV, MediaPipe, DJI Tello SDK*

- Developed a real-time computer vision system using MediaPipe and OpenCV to detect and map hand landmarks, translating ASL-inspired gestures into structured drone control signals.
- Integrated DJI Tello SDK to execute low-latency flight commands (takeoff, movement, rotation, landing) enabling hands-free drone navigation via gesture recognition.